

RESEARCH OVERVIEW

I develop stable visual intelligence for dynamic worlds, connecting perception, understanding, and generation. My research studies how to preserve object identity, spatial relationships, and temporal consistency as observations and scenes change, with applications in autonomous systems, digital humans, and healthcare.

- AI for Digital Twins** 2025 - Present
 - World Models; Digital Humans
- AI for Autonomous Mobility** 2020 - Present
 - 3D Occupancy Prediction; Open-Vocabulary/Cross-Domain/Open-Set Object Detection
- AI for Scientific Innovation** 2024 - 2025
 - Nanophotonics; Meta-Optics; Graph-Based Learning
- AI for Transforming Medicine** 2020 - 2024
 - MRI Brain Analysis; Medical Report Generation; 4D Surgical Simulation

PROFILE HIGHLIGHTS

- Academic & Industrial Perspective**
 - Founded a seed-funded Hong Kong startup while pursuing top-tier research (50+ papers; 15+ lead-author papers).
- Large-Scale GPU Experimentation**
 - Extensive hands-on experience with large-scale GPU clusters (10,000+ GH200).
- Leadership & Collaboration**
 - Led a Swiss national AI initiative with 15 leading PIs; NeurIPS workshop organizer.
- Rapid Citation Impact**
 - U-KAN (co-first author) received 600+ citations within 1.5 years.
- Open-Source Impact**
 - Stable Video Infinity attracted **2.6k+** GitHub stars and was deployed on the Poe platform.

RESEARCH HIGHLIGHTS

- Oral/Spotlight/Highlight Presentations (First Author)**
 - AAAI 2022; CVPR 2022; ICCV 2023; ECCV 2024; ICCV 2025; NeurIPS 2025; ICLR 2026
- Best Paper Finalist (First Author)**
 - CVPR 2022 (33/8161)
- Full-Score Submission (First Author)**
 - ICCV 2025, final scores: 6, 6, 6
- Cover Paper (Second Author)**
 - ACS Photonics, 2024
- AAAI 2025 Top-1 Influential Paper (Co-First Author)**
 - U-KAN: medical image segmentation and generation backbone

WORK EXPERIENCE

- Ant Group** Jun. 16, 2026 - Present
 - Staff Research Engineer
 - Research areas: multimodal intelligence, world models, and AI for digital twins
- École Polytechnique Fédérale de Lausanne (EPFL)** Feb. 2025 - Jun. 2026
 - Research Scientist, Robotics and Computer Vision
 - Supervisor:** Prof. Alexandre Alahi (advised by Prof. Fei-Fei Li at Stanford)
- The Chinese University of Hong Kong (CUHK)** Mar. 2024 - Feb. 2025
 - Postdoctoral Researcher, Electronic Engineering
 - Supervisor:** Prof. Yixuan Yuan
- ScholaGO Education Technology Company Limited, Hong Kong** 2024 - Present
 - Founder & CEO
 - Developed multimodal educational products; secured seed funding from HK Tech 300 and HKSTP.

EDUCATION

- City University of Hong Kong (CityU)** Sep. 2020 - Feb. 2024
 - Ph.D., Electrical Engineering
 - 3.5 years; early graduation
 - Supervisor:** Prof. Yixuan Yuan
- Tianjin University (TJU)** Sep. 2016 - Jun. 2020
 - B.S., Electrical Engineering
 - GPA: 3.83/4.00 (Top 2%)

1. [Generation] EverAnimate: Minute-Scale Human Animation via Latent Flow Restoration
Wuyang Li, Yang Gao, Mariam Hassan, Lan Feng, Wentao Pan, Po-Chien Luan, Alexandre Alahi
Conference on Neural Information Processing Systems (**NeurIPS**), 2026
2. [Generation] SurgVista: Long-Horizon Surgical World Modeling with Plausible Instrument-Tissue Dynamics
Wentao Pan*, **Wuyang Li**^{†,*}, Shengyuan Liu, Xinyu Liu, Hengyu Liu, Yixuan Yuan
Conference on Neural Information Processing Systems (**NeurIPS**), 2026
3. [Generation] Drift-Resistant Navigation World Model with Anchored Epipolar Guidance
Po-Chien Luan, Zimin Xia, **Wuyang Li**, Yang Gao, Alexandre Alahi
Conference on Neural Information Processing Systems (**NeurIPS**), 2026
4. [Generation] Proprio: Latent Self-Scoring and Inference-Time Refinement for Physically Plausible Video Generation
Mariam Hassan, Kaouther Messaoud, **Wuyang Li**, Alexandre Alahi
Conference on Neural Information Processing Systems (**NeurIPS**), 2026
5. [Generation] Anchored Video Generation: Decoupling Scene Construction and Temporal Synthesis in Text-to-Video Diffusion Models
Mariam Hassan, Bastien Van Delft, **Wuyang Li**[†], Alexandre Alahi
European Conference on Computer Vision (**ECCV**), 2026
6. [Understanding] Social-Mamba: Socially-Aware Trajectory Forecasting with State-Space Models
Po-Chien Luan, **Wuyang Li**[†], Yang Gao, Alexandre Alahi
European Conference on Computer Vision (**ECCV**), 2026
7. [Generation] Stable Video Infinity: Infinite-Length Video Generation with Error Recycling
Wuyang Li, Wentao Pan, Po-Chien Luan, Yang Gao, Alexandre Alahi
International Conference on Learning Representations (**ICLR**), 2026, Oral (1.3%)
8. [Perception] Deformable Gaussian Occupancy: Decoupling Rigid and Nonrigid Motion with Factorized Distillation
Yang Gao, **Wuyang Li**[†], Po-Chien Luan, Alexandre Alahi
IEEE Conference on Computer Vision and Pattern Recognition (**CVPR**), 2026
9. [Perception] VoxDet: Rethinking 3D Semantic Occupancy Prediction as Dense Object Detection
Wuyang Li, Zhu Yu, Alexandre Alahi
Conference on Neural Information Processing Systems (**NeurIPS**), 2025, Spotlight (3%)
10. [Perception] MetaScope: Optics-Driven Neural Network for Ultra-Micro Metalens Endoscopy
Wuyang Li^{*}, Wentao Pan^{*}, Xiaoyuan Liu^{*}, Zhendong Luo, Chenxin Li, Hengyu Liu, Din Ping Tsai, Mu Ku Chen, Yixuan Yuan
IEEE International Conference on Computer Vision (**ICCV**), 2025, Final review scores (6, 6, 6)
11. [Perception] U-KAN Makes a Strong Backbone for Medical Image Segmentation and Generation
Chenxin Li^{*}, Xinyu Liu^{*}, **Wuyang Li**^{*}, Cheng Wang^{*}, Hengyu Liu, Yifan Liu, Zhen Chen, Yixuan Yuan
AAAI Conference on Artificial Intelligence (**AAAI**), 2025, AAAI 2025 Top-1 Influential Paper
12. [Perception] CLIFF: Continual Latent Diffusion for Open-Vocabulary Object Detection
Wuyang Li, Xinyu Liu, Jiayi Ma, Yixuan Yuan
European Conference on Computer Vision (**ECCV**), 2024, Oral (1.8%)
13. [Generation] Boosting Medical Image Analysis via Motion-Informed Generative Videos
Wuyang Li, Xinyu Liu, Qiushi Yang, Yixuan Yuan
Medical Image Computing and Computer Assisted Intervention (**MICCAI**), 2024
14. [Perception] Novel Scenes & Classes: Towards Adaptive Open-set Object Detection
Wuyang Li, Xiaoqing Guo, Yixuan Yuan
IEEE International Conference on Computer Vision (**ICCV**), 2023, Oral (1.8%)
15. [Perception] Adjustment and Alignment for Unbiased Open Set Domain Adaptation
Wuyang Li, Jie Liu, Bo Han, Yixuan Yuan
IEEE Conference on Computer Vision and Pattern Recognition (**CVPR**), 2023
16. [Perception] SIGMA++: Improved Semantic-complete Graph Matching for Domain Adaptive Object Detection
Wuyang Li, Xinyu Liu, Yixuan Yuan
IEEE Transactions on Pattern Analysis and Machine Intelligence (**TPAMI**), 2023
17. [Perception] SIGMA: Semantic-complete Graph Matching for Domain Adaptive Object Detection
Wuyang Li, Xinyu Liu, Yixuan Yuan
IEEE Conference on Computer Vision and Pattern Recognition (**CVPR**), 2022, Best Paper Finalist (0.4%)

18. [\[Perception\]](#) SCAN: Cross Domain Object Detection with Semantic Conditioned Adaptation
Wuyang Li, Xinyu Liu, Xiwen Yao, Yixuan Yuan
AAAI Conference on Artificial Intelligence (AAAI), 2022, [Oral \(4.5%\)](#)
- [\[Generation\]](#) Synthesizing Realistic fMRI: A Physiological Dynamics-Driven Hierarchical Diffusion Model for Efficient
19. fMRI Acquisition
Yufan Hu, Yu Jiang, **Wuyang Li**, Yixuan Yuan
International Conference on Learning Representations (ICLR), 2025
- [\[Generation\]](#) InstantSplamp: Fast and Generalizable Stenography Framework for Generative Gaussian Splatting
20. Chenxin Li*, Hengyu Liu*, Zhiwen Fan, **Wuyang Li**, Yifan Liu, Panwang Pan, Yixuan Yuan
International Conference on Learning Representations (ICLR), 2025
- [\[Perception\]](#) Universal Domain Adaptive Object Detection via Dual Probabilistic Alignment
21. Yuanfan Zheng*, Jinlin Wu*, **Wuyang Li**, Zhen Chen
AAAI Conference on Artificial Intelligence (AAAI), 2025
- [\[Perception\]](#) Stereo Vision Meta-Lens-Assisted Driving Vision
22. Xiaoyuan Liu, **Wuyang Li**, Takeshi Yamaguchi, Zihan Geng, Takuo Tanaka, Din Ping Tsai, Mu Ku Chen
ACS Photonics, 2024, [Selected as Cover Paper](#)
- [\[Perception\]](#) Flaws can be Applause: Unleashing Potential of Segmenting Ambiguous Objects in SAM
23. Chenxin Li*, Yuzhi Huang*, **Wuyang Li**, Hengyu Liu, Xinyu Liu, Qing Xu, Zhen Chen, Yue Huang, Yixuan Yuan
Conference on Neural Information Processing Systems (NeurIPS), 2024
- [\[Perception\]](#) When 3D Partial Points Meets SAM: Tooth Point Cloud Segmentation with Sparse Labels
24. Yifan Liu, **Wuyang Li**, Cheng Wang, Hui Chen, Yixuan Yuan
Medical Image Computing and Computer Assisted Intervention (MICCAI), 2024, [Early Accept \(14.0%\)](#)
- [\[Perception\]](#) Latent Diffusion Label Rectification for Semi-supervised Medical Image Segmentation
25. Xinyu Liu, **Wuyang Li**, Yixuan Yuan
Medical Image Computing and Computer Assisted Intervention (MICCAI), 2024
- [\[Understanding\]](#) fMRI to T1w MRI Knowledge Transfer Network for Brain Multi-phenotype Prediction
26. Zhibin He, **Wuyang Li**, Yu Jiang, Zhihao Peng, Pengyu Wang, Xiang Li, Tianming Liu, Junwei Han, et al.
Medical Image Computing and Computer Assisted Intervention (MICCAI), 2024
- [\[Perception\]](#) LGS: A Light-weight 4D Gaussian Splatting for Efficient Surgical Scene Reconstruction
27. Hengyu Liu*, Yifan Liu*, Chenxin Li*, **Wuyang Li**, Yixuan Yuan
Medical Image Computing and Computer Assisted Intervention (MICCAI), 2024
- [\[Generation\]](#) Endora: Video Generation Models as Endoscopy Simulators
28. Chenxin Li, Hengyu Liu, Yifan Liu, Brandon Feng, **Wuyang Li**, Xinyu Liu, Zhen Chen, Jing Shao, Yixuan Yuan
Medical Image Computing and Computer Assisted Intervention (MICCAI), 2024
- [\[Understanding\]](#) Foundation Model for Any Phenotype Prediction via fMRI to sMRI Knowledge Transfer
29. Zhibin He, **Wuyang Li**, Yifan Liu, Xinyu Liu, Junwei Han, Tuo Zhang, Yixuan Yuan
IEEE Transactions on Medical Imaging (TMI), 2024
- [\[Understanding\]](#) MRM: Masked Relation Modeling for Medical Image Pre-Training with Genetics
30. Qiushi Yang, **Wuyang Li**, Baopu Li, Yixuan Yuan
IEEE International Conference on Computer Vision (ICCV), 2023
- [\[Perception\]](#) Intervention & Interaction Federated Abnormality Detection with Noisy Clients
31. Xinyu Liu, **Wuyang Li**, Yixuan Yuan
Medical Image Computing and Computer Assisted Intervention (MICCAI), 2022, [Oral](#), [Early Accept](#)
- [\[Perception\]](#) Towards Robust Adaptive Object Detection under Noisy Annotations
32. Xinyu Liu, **Wuyang Li**, Qiushi Yang, Baopu Li, Yixuan Yuan
IEEE Conference on Computer Vision and Pattern Recognition (CVPR), 2022
- [\[Perception\]](#) HTD: Heterogeneous Task Decoupling for Two-Stage Object Detection
33. **Wuyang Li**, Zhen Chen, Baopu Li, Dingwen Zhang, Yixuan Yuan
IEEE Transactions on Image Processing (TIP), 2021

AWARDS

• Outstanding Academic Performance Award, CityU	2022 - 2023
• Outstanding research achievements	
• Research Tuition Scholarship, CityU	2022 - 2023
• Outstanding research achievements	
• First Place in the Polyp Detection Challenge	2021
• 3rd International Endoscopy Computer Vision Challenge and Workshop (EndoCV2021)	
• Undergraduate National Scholarship	2017 - 2018
• The highest undergraduate honor for a student in China	
• Silver Award in Tianjin Advanced Mathematical Competition	2017
• Selected to represent Tianjin University	

SERVICE

• Swiss AI Initiative Project Lead [Link]	2025
• Led a national project with 15 leading Swiss PIs; gained large-scale GPU cluster experience (10,000+ GH200).	
• NeurIPS 2024 Workshop Organizer [Link]	2024
• Organized the Advancements in Multimodal Medical Foundation Models workshop	
• Research Funding Proposals	2022 - 2024
• Assisted my advisor in preparing several competitive research funding proposals	
• Startup Funding [Link]	2023
• Secured seed funding from HK Tech 300 and HKSTP	
• Founder, ScholaGO Education Technology Company Limited, Hong Kong [Link]	2024
• Founded a company developing multimodal educational products	
• Conference and Journal Reviewer	
• ICLR/ICML/NeurIPS; CVPR/ICCV; TPAMI; IJCV; TIP; NeurIPS Top Reviewer	